Document Similarity and Retrieval using the VSM

A. With Sparse Embeddings

- Use the News Category Dataset available on Kaggle:
  https://www.kaggle.com/datasets/rmisra/news-category-dataset
- This dataset contains ~200k news headlines from HuffPost, classified into categories such as politics, tech, and sports.

- Setup and Preprocessing
  - Load the dataset and randomly select a subset of headline texts (e.g., 10,000).
  - Preprocess the text through tokenization, lowercasing, stopwords and punctuation removal, lemmatization.

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- Vector Space Model representation
  - Convert the preprocessed documents to TF-IDF vectors (use `TfidfVectorizer` from scikit-learn)
  - Store the resulting matrix and vocabulary
- Document Retrieval
  - Prompt the user for a query (e.g., "government budget")
  - Preprocess the query using the same pipeline
  - Convert the query to a TF-IDF vector using the same vectorizer
  - Compute the cosine similarity between the query and all documents
  - Return and display the top 5 most relevant documents



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- Analysis
  - Manually inspect the top results: do they seem relevant?
  - Try at least 10 queries from different categories and observe the results

